# Mikhail Trunov

## Contact

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## Summary

AI/ML engineer with a full-stack background. Currently developing a machine-learning classifier for ultrasound images at ASTRAIA, on top of years of building production medical software in Python. I use AI agents daily to design, build and ship products end to end.

## Skills

- AI & ML: PyTorch (primary), image classification, medical imaging, LLM pipelines (Gemini), TensorFlow, LangChain, MCP
- AI-assisted development: Claude Code, Codex, OpenClaw (daily use)
- Languages: Python (expert), JavaScript, TypeScript, Rust (beginner)
- Web Frameworks: Django, FastAPI, React, Next.js, expo
- DevOps & Cloud: Docker, Ansible, Terraform, CI/CD, GCP
- Testing: pytest, unittest, jest
- Databases: PostgreSQL, SQLite, Redis
- Other Tools: Git, Linux, REST APIs, OpenAPI/Swagger, Telegram bots

## Built by me

### KinderCafe.app

Smartphone app for finding child-friendly locations and events in Germany.

- Built an LLM pipeline with Gemini that parses events from the websites of family and community centres
- Tech stack: Expo, Firebase, Cloud Run, Wagtail backend with PostGIS, Gemini

## Experience

### Medical Developer – NEXUS / ASTRAIA, Ismaning, Germany

*Nov 2021 – Present (5 years)*

- Developing a machine-learning classifier for ultrasound images
- Designed and implemented the FMF (Fetal Medicine Foundation) risk algorithm in Python to assess pregnancy complication risks
- Developed a Django-based web application to host and deliver the FMF algorithm
- Built internal tools using Tauri to support the design and development of the main medical software
- Migrated CI/CD pipelines from Jenkins to Bitbucket Pipelines, improving automation and maintainability

## Education

2016 - 2019 **M.Sc. in Advanced Materials and Processes (MAP)** – FAU Erlangen-Nürnberg, Germany

2010 - 2012 **M.Eng. in Materials Science** – Volgograd State Technical University, Russia

2006 - 2010 **B.Eng. in Industrial Engineering** – Volgograd State Technical University, Russia

## Languages

- English (fluent)
- German (intermediate)
- Russian (Native)